Astronomical Object Classification Using Machine Learning

Abstract

With the rapid growth of astronomical data from large sky surveys, manually classifying celestial objects has become increasingly difficult. This project explores how machine learning can be used to classify objects such as Stars, Galaxies, and Quasi-Stellar Objects (QSOs). Using photometric data from the Sloan Digital Sky Survey (SDSS), a Random Forest model was trained on both raw features and engineered features like color indices and redshift. The model achieved an accuracy of around 97.6%, showing that combining physics-based understanding with machine learning can give strong results. The final model was also deployed as a web application, making it interactive and easy to use.

Keywords

Astronomy, Machine Learning, SDSS, Classification, Random Forest, Redshift, Photometry

Introduction

Astronomy today is heavily driven by data. Surveys like the Sloan Digital Sky Survey (SDSS) collect massive amounts of information about celestial objects. Because of this, manually identifying and classifying these objects is no longer practical.

Machine learning offers a powerful solution to this problem. By learning patterns from data, models can automatically classify objects based on their properties.

In this project, the goal was to:

- Classify celestial objects into Stars, Galaxies, and QSOs
- Understand how physical features like redshift and light properties affect classification

- Build and deploy a working machine learning system

Dataset Description

The dataset used in this project comes from the **Sloan Digital Sky Survey (SDSS)**. It includes observations of different celestial objects along with their physical properties.

- ## Main Features

- **u, g, r, i, z**: These represent brightness in different wavelength bands
- **Redshift**: Gives an idea of how far the object is
- **Class**: The actual type of object (STAR, GALAXY, QSO)

- ## Data Preprocessing

Before training the model, the data needed some cleaning:

- Removed invalid or extreme values
- Filtered out noisy data
- Ensured all values were within reasonable ranges

This step was important to make sure the model learns from good-quality data.

Feature Engineering

Instead of only using raw data, new features were created to improve performance.

- ## Colour Indices

Colour indices were calculated by subtracting magnitudes:

- $g-\mathrm{r}$
- $u-\mathrm{g}$
- $r-\mathrm{i}$
- $i-\mathrm{z}$
- $u-\mathrm{r}$

These values help capture how objects emit light across different wavelengths, which is very useful in distinguishing between stars, galaxies, and QSOs.

4. Methodology

○ Model Selection

A **Random Forest Classifier** was chosen for this project. It works well because:

- It handles complex relationships in data
- It is resistant to noise
- It usually gives strong accuracy without heavy tuning

○ Training Process

- The dataset was split into training and testing sets
- The model was trained using selected features
- Default parameters were used initially

○ Evaluation Metrics

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

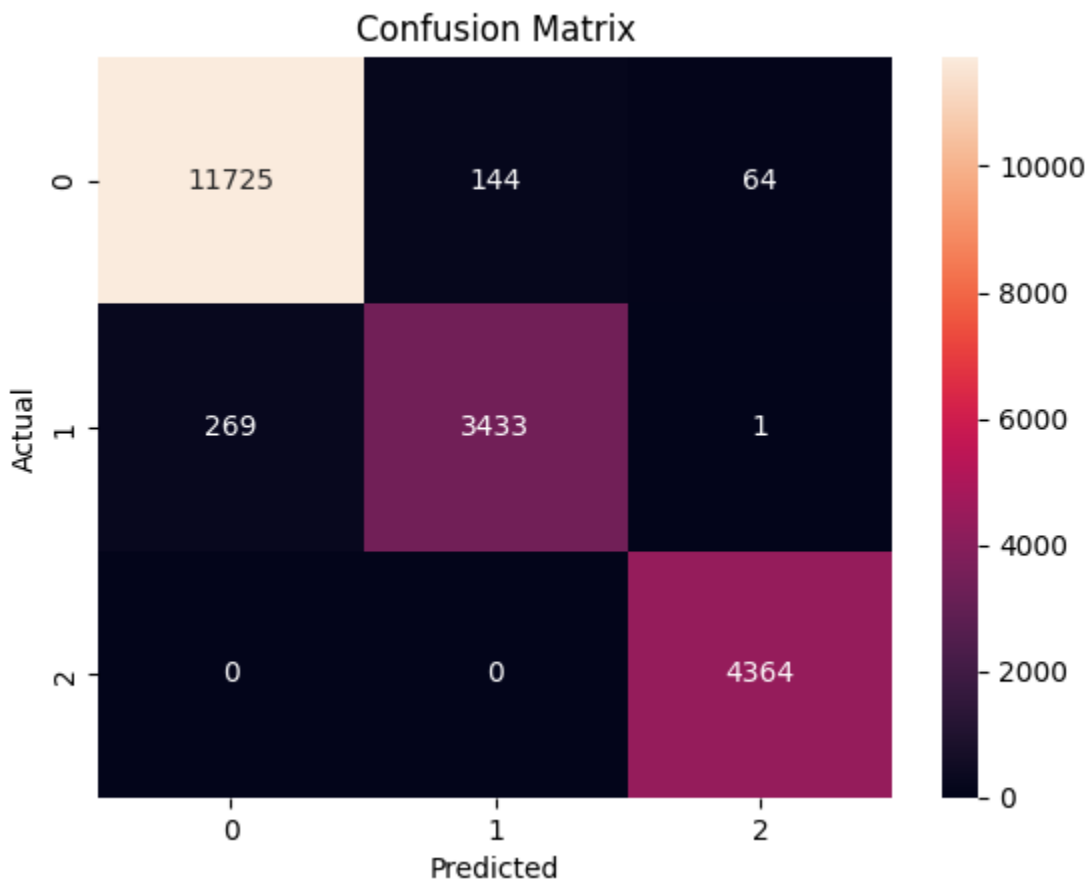
These metrics help understand how well the model performs across different classes.

Results

The model achieved an accuracy of around **97.6%**, which is quite strong.

Visualization & Analysis

- Confusion matrix

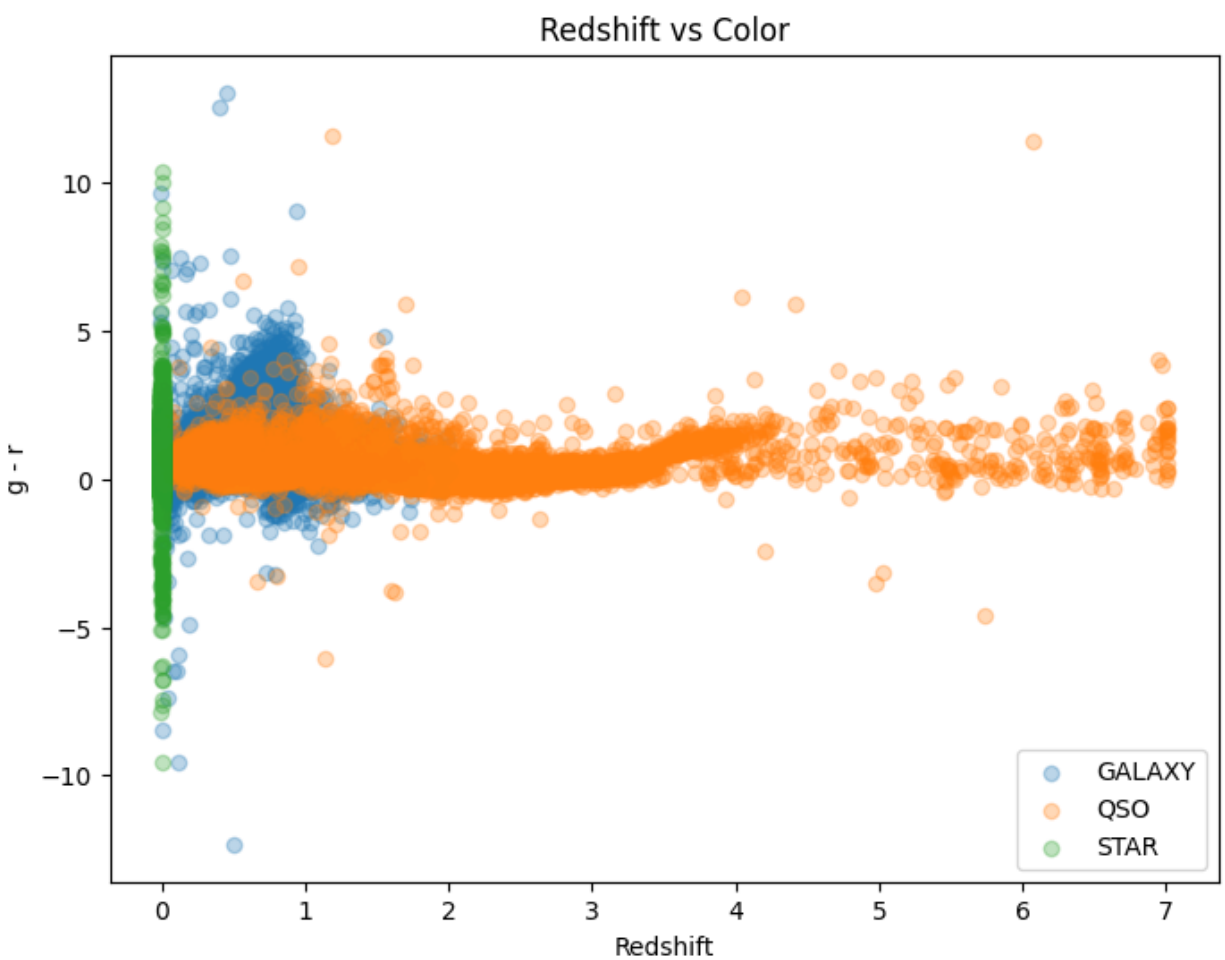



```
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

: - Confusion Matrix showing classification performance across STAR, GALAXY, and QSO classes. The model performs strongly overall, with minor confusion between QSOs and galaxies.

- Colour Scatter Plot

```
import matplotlib.pyplot as plt

colors = {
    "STAR": "blue",
    "GALAXY": "green",
    "QSO": "red"
}

plt.figure(figsize=(8,6))

for obj in df_clean["class"].unique():
    subset = df_clean[df_clean["class"] == obj]
    plt.scatter(subset["redshift"], subset["g_r"],
                label=obj, alpha=0.3)

plt.xlabel("Redshift")
plt.ylabel("g - r")
plt.title("Redshift vs Color")
plt.legend()

plt.show()
```



: - Scatter plot of g vs r magnitudes showing distribution of different celestial object classes. Distinct clustering patterns can be observed.

- Feature importance

```
importance = model.feature_importances_

import pandas as pd

df_imp = pd.DataFrame({
    "feature": X.columns,
    "importance": importance
}).sort_values(by="importance", ascending=False)

print(df_imp)
```

[57]

```
...          feature  importance
    13     redshift    0.436865
    19          r_i    0.095074
    21          u_r    0.090678
    17          g_r    0.070421
    18          u_g    0.053659
    20          i_z    0.046588
    7             z    0.036015
    4             g    0.027621
    5             r    0.020620
    14        plate    0.020305
    6             i    0.020072
    12  spec_obj_ID    0.019727
    3             u    0.018577
    15          MJD    0.015818
    1         alpha    0.005449
    2         delta    0.005331
    16     fiber_ID    0.004419
    11     field_ID    0.004186
    8        run_ID    0.003670
    0        obj_ID    0.003129
    10      cam_col    0.001774
    9      rerun_ID    0.000000
```

: - The feature importance analysis reveals that **redshift is the most dominant feature**, with an importance score of approximately **0.43**, significantly higher than all other features.

This indicates that redshift plays a critical role in distinguishing between different astronomical objects, particularly **Quasi-Stellar Objects (QSOs)**, which are typically located at much greater distances compared to stars and galaxies.

Colour index features such as:

- $r-$i
- $u-$r
- $g-$r

also contribute significantly to the model. These features represent differences in light intensity across wavelengths and help in distinguishing between **stellar and galactic objects**.

On the other hand, raw features like individual magnitudes (**u, g, r, i, z**) and metadata (e.g., plate, fiber_ID) have relatively lower importance, suggesting that **derived features are more informative than raw measurements**.

Observations

- **Stars** were classified very accurately
- **Galaxies** also showed high accuracy
- **QSOs** were slightly harder to classify

This is expected because QSOs and galaxies sometimes share similar properties.

Discussion

The results show that machine learning can effectively classify astronomical objects when meaningful features are used.

Key Insights

- **Redshift** is extremely useful for identifying QSOs
- **Color indices** help separate stars and galaxies
- Some errors happen because certain objects naturally overlap in their properties

This means the limitation is not just in the model, but also in the nature of the data.

Deployment

To make the model usable, it was deployed as a web application using **Gradio**.

Features of the App

- Users can enter photometric values
- The model gives predictions instantly
- Confidence score is displayed
- Simple and clean interface

This step makes the project practical, not just theoretical.

Conclusion

This project shows that machine learning can be successfully used in astronomy for object classification. By combining scientific understanding with data-driven methods, high accuracy can be achieved.

The deployment of the model further demonstrates its real-world usability.

Future Work

There are several ways this project can be improved:

- Using deep learning models
- Adding more features or spectral data
- Improving classification of QSOs
- Scaling to larger datasets

Acknowledgment

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